

News

OH Upson 447

Today 12-2 w/TA.

Energy & Work

- Review Momentum
- Energy
- Work
- Conservative Forces

HW3

Due Friday 6pm.

Questions on Canvas

- Gradescope & Matlab

Autograder will open today

ReviewConservation of
Linear Momentum

LMB

$$\sum \vec{F}_{\text{EXT}} = \dot{\vec{p}}$$

$$\text{w/ } \dot{\vec{p}} = 0$$

$$\vec{p} = \text{const.}$$

Conservation of
Angular Momentum

AMB

$$\sum \vec{M}_{\text{EXT}/O} = \dot{\vec{H}}_O$$

$$\text{w/ } \dot{\vec{H}}_O = 0$$

$$\vec{H}_O = \text{const.}$$

Energy

$$\vec{F} = m\vec{a}$$

Dot product

$$\vec{F} \cdot \vec{v} = m\vec{a} \cdot \vec{v}$$

$$P = \frac{d}{dt} \left(\underbrace{\frac{m\vec{v}^2}{2}}_{E_k} \right)$$

$$P = \dot{E}_k$$

Aside

$$\vec{v}^2 = v_x^2 + v_y^2 + v_z^2$$

$$= \vec{v} \cdot \vec{v}$$

$$\frac{d}{dt}(\vec{v}^2) = \vec{v} \cdot \dot{\vec{v}} + \dot{\vec{v}} \cdot \vec{v}$$

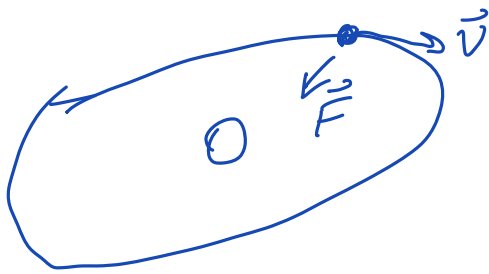
$$= 2\vec{v} \cdot \dot{\vec{v}}$$

$$= 2\vec{v} \cdot \vec{a}$$

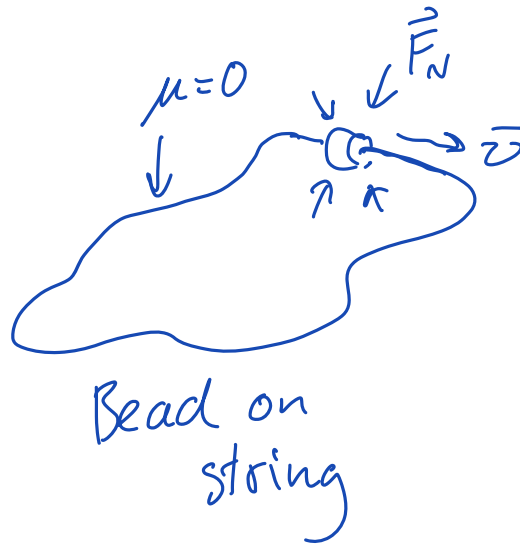
Special Case

$$\text{If } \vec{F} \perp \vec{v}, \text{ then } \vec{F} \cdot \vec{v} = 0$$

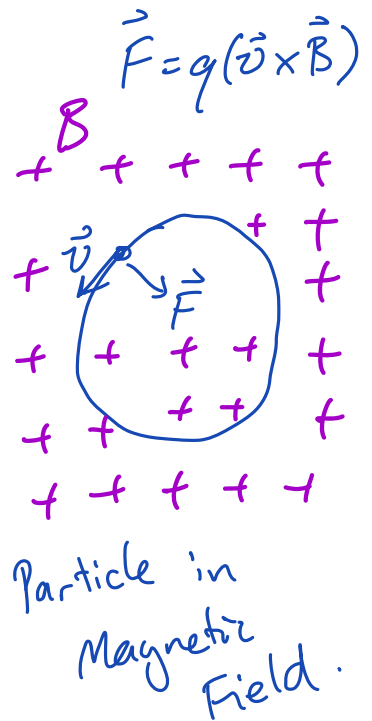
If the force is normal to the velocity
no energy is transferred.

Examples

Circular Motion



Bead on string



Particle in Magnetic Field.

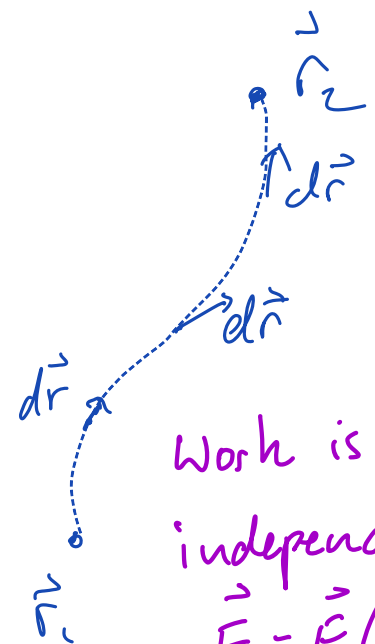
Work

Integrate P

$$\int_{t_1}^{t_2} P \, dt = \int_{t_1}^{t_2} \dot{E}_k \, dt$$

$$\int_{t_1}^{t_2} \vec{F} \cdot \underbrace{\vec{v} dt}_{d\vec{r}} = \underbrace{\Delta E_k}_W$$

$$\boxed{\int_{r_1}^{r_2} \vec{F} \cdot d\vec{r} = W}$$



Work is path independent

$$\vec{F} = \vec{F}(\vec{r})$$

(not $\vec{F}(\vec{t}, \vec{v}, \vec{r})$)

Conservative Forces

Yes

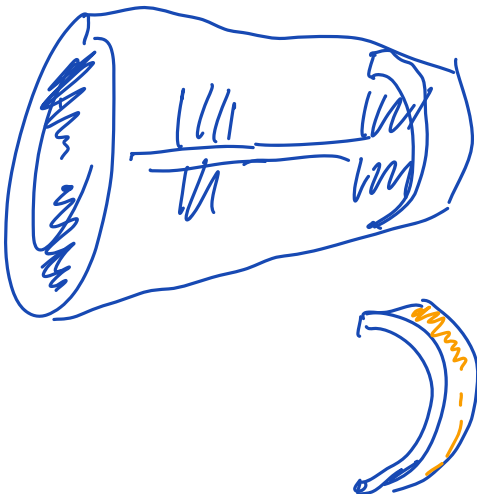
Gravity

Electrostatic

Centripetal

Springs

Buoyancy



No

Friction

Drag

Mechanical Pumping

Viscosity

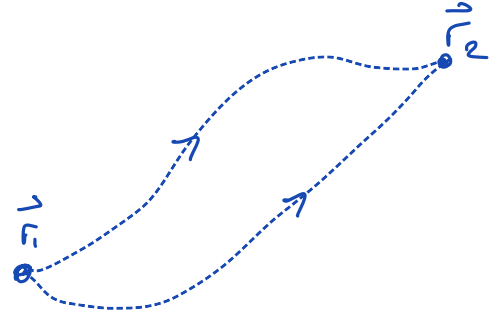
Electrical Resistance



Dissipative
Forces

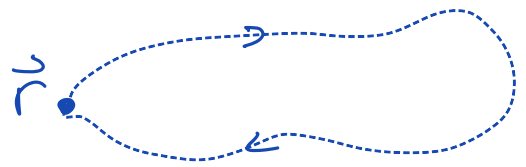
Analogous properties of Conservative forces

$$i) \int_{r_1}^{r_2} \vec{F}(\vec{r}) \cdot d\vec{r} = W$$



$$ii) \oint \vec{F} \cdot d\vec{r} = 0$$

within a simply connected region



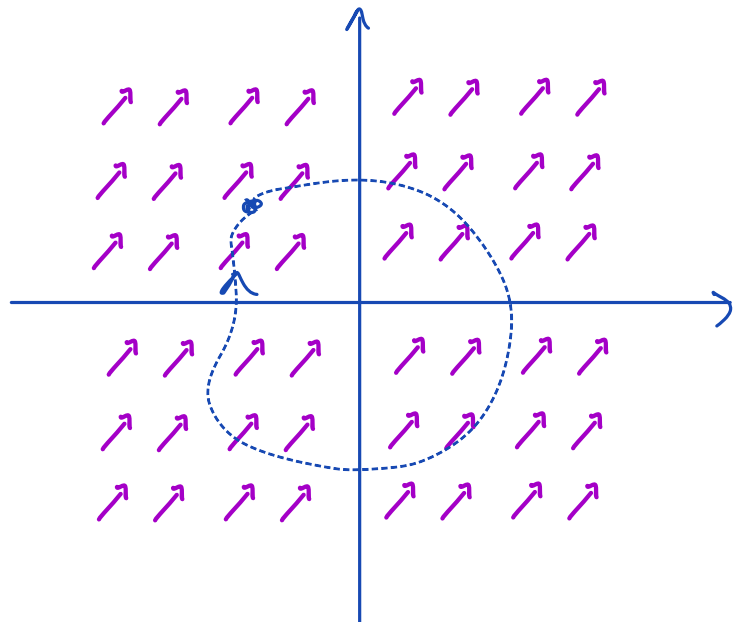
$$iii) \vec{\nabla} \times \vec{F} = 0$$

'Curl of vector field is zero'

$$3D: 0 = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \hat{x} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \hat{y} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \hat{z}$$

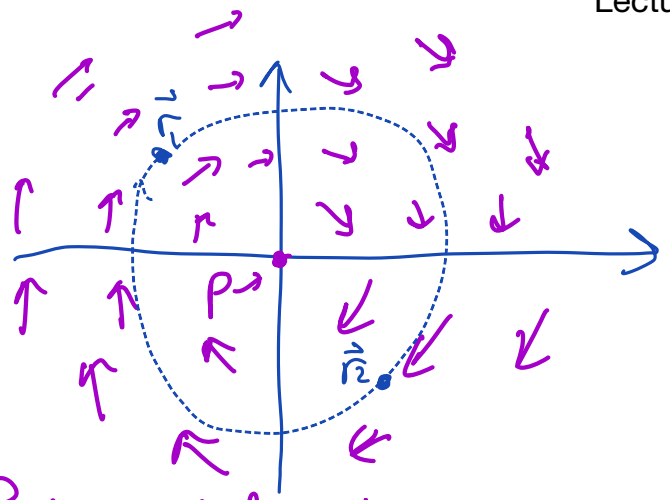
$$2D: 0 = \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}$$

$$iv) \vec{F} = -\vec{\nabla} E_p$$



If one of these is true, they are all true.

Example
of Non-conservative
Force Field.
'The Whirlpool'



Field at P is undefined.

- i) Path from $\vec{r}_1$ to $\vec{r}_2$ will experience different W if clockwise or anti-clockwise
- ii) Path around field $\vec{r}_1$ to $\vec{r}_1$ will experience some W / will encompass point P which is a singularity / not simply connected.
- iii) Field tends to trend clockwise / $\vec{\nabla} \times \vec{F} \neq 0$
- iv) The force experienced at any point in the field is not explicitly due to the gradient of the potential energy.

